Weierstrass Convergence Theorem.

Def. A sequence $f_n : G \to \mathbb{C}$ converges locally uniformly to $f$ if:
for every Compact $K \subset G$,

$$\sup_{z \in K} |f_n(z) - f(z)| \to 0 \quad \text{as } n \to \infty.$$

Thm. Suppose that $f_n : G \to \mathbb{C}$ is holomorphic on $G$ and $f_n \to f$ locally uniformly, then $f$ is holomorphic. Moreover, $f_n' \to f'$ locally uniformly on $G$. ($G$ open)

Proof. Given $z_0 \in G$, take $\delta > 0$ such that $\overline{B(z_0, r)} \subset G$. By the hypothesis, $f_n \to f$ uniformly on $\overline{B_\delta(z_0)}$, and so $f$ is Continuous on $B_\delta(z_0)$.
Let $T$ be a triangular path inside $B_\delta(z_0)$. Since $f_n \to f$ uniformly,

$$\int_T f_n(z)\, dz \to \int_T f(z)\, dz.$$

But, by the Cauchy-Goursat's theorem, each $n \in \mathbb{N}$, $\int_T f_n(z)\, dz = 0$, so $\int_T f(z)\, dz = 0$.

Now, by the Morera's theorem, $f$ is Holomorphic on $B_\delta(z_0)$.
Since $z_0$ was arbitrary, so $f$ is Holomorphic on $G$. To show the second assertion, let $K \subset G$ be a Compact Subset. Since $\mathbb{C}$ is $T_4$ space and $G$ is open, there is an open $U$ such that $K \subseteq U \subseteq \overline{U} \subseteq G$.
Now, for each $\delta > 0$, define

$$U_\delta := \{ z \in U \mid \overline{D_\delta(z)} \subset U \} \text{ where } D_\delta = \{ z' \in \mathbb{C} \mid |z' - z| < \delta \}.$$

Given $z \in U_\delta$, by the Cauchy's integral formula,

$$f'(z) - f_n'(z) = \frac{1}{2\pi i} \cdot \int_{C_\delta} \frac{f(u) - f_n(u)}{(u - z)^2}\, du$$

where $C_\delta$ is a curve centered at $z$ with radius $\delta > 0$. Therefore,

$$|f'(z) - f_n'(z)| \leq \frac{1}{2\pi} \cdot \int_{C_\delta} \frac{|f(u) - f_n(u)|}{|u - z|^2} |du| \leq \frac{1}{2\pi} \cdot \frac{2\pi\delta}{\delta^2} \cdot \sup_{u \in C_\delta} |f(u) - f_n(u)|$$

$$\leq \frac{1}{\delta} \cdot \sup_{u \in \overline{U}} |f(u) - f_n(u)| \to 0 \text{ as } n \to \infty.$$

Since the bound that tends to zero is independent of choice of $z$,
Therefore, we can choose $\delta > 0$ small enough to $K \subset U_\delta$,
so $f_n' \to f'$ uniformly. ☒